

Joseph Zhang

FULL STACK ENGINEER

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SUMMARY

AI Product Engineer and mobile application developer with 8+ years of experience building production software, LLM-powered products, and customer-facing mobile applications. Experienced in React Native and native iOS development using Swift, SwiftUI, and UIKit, including mobile API integration, asynchronous workflows, background processing, push notifications, subscriptions, analytics, and App Store releases. Launched 30+ independent iOS and macOS applications with more than 4 million installs. Also experienced in building RAG pipelines, AI agents, semantic search, structured-output workflows, and full-stack AI applications using TypeScript, Python, React, Next.js, Node.js, FastAPI, PostgreSQL, OpenAI and Gemini.

TECHNICAL PROFICIENCIES

Languages: Python, TypeScript, JavaScript, Swift, C#, C++, Rust, Scala, Elixir

AI / ML: OpenAI API, LangChain, LangGraph, Hugging Face Transformers, PyTorch, TensorFlow, RAG, AI Agents, Agentic Workflows, Embeddings, Semantic Search, Tool Calling, Vector Search, Function Calling, Structured Outputs, Prompt Engineering

Frontend: React, Next.js

Mobile: React Native, iOS, SwiftUI, UIKit, Android, Kotlin, Push Notifications, In-App Purchases, App Store Connect, TestFlight

Backend: Node.js, FastAPI, Django, REST APIs, GraphQL, Background Jobs, Asynchronous Processing

Database: PostgreSQL, MySQL, Microsoft SQL Server, Redis, Firebase, pgvector, FAISS

Cloud & DevOps: AWS, AWS Lambda, GCP, Microsoft Azure, Docker, CI/CD, GitHub Actions, Git, Linux

Testing: Evaluation Harnesses, Regression Evaluation, Retrieval-Quality Testing, A/B Testing, Structured Logging, Monitoring, Crash Analytics

EXPERIENCE

Senior Software Engineer / AI Engineer · First Orion, North Little Rock

OCT 2019 — PRESENT

- Led the development of production software and AI systems for caller identity, reputation scoring, spam detection, and anomaly detection across high-volume telecommunications data.
- Built end-to-end product features across React and TypeScript frontends, backend APIs, data-processing services, databases, and cloud infrastructure.
- Developed responsive web interfaces and internal analytics tools using React, TypeScript, reusable components, state management, and REST API integrations.
- Designed and implemented backend services using Python, Node.js, REST APIs, PostgreSQL, Redis, and asynchronous processing to support mobile, web, and AI-powered product workflows.
- Developed customer-facing mobile features using React Native, Swift, UIKit, Kotlin, and Jetpack Compose, integrating caller identity and reputation intelligence into iOS and Android applications.
- Built and maintained native mobile bridges for iOS and Android, enabling React Native applications to access platform-specific SDKs, device APIs, native modules, and background services.
- Integrated React Native, iOS, Android, and web applications with backend APIs, authentication services, analytics, push notifications, data services, and production monitoring platforms.
- Managed iOS and Android release workflows, including build configuration, code signing, provisioning, testing, TestFlight and internal distribution, App Store and Google Play submissions, release validation, crash investigation, and post-launch monitoring.
- Built ingestion, enrichment, and feature-processing pipelines combining call metadata, spam reports, carrier signals, engagement history, and behavioral patterns for ranking and classification workflows.

- Developed real-time Python scoring services using historical behavior, frequency signals, confidence thresholds, and reputation indicators to identify high-confidence caller identities and suspicious activity.
- Implemented NLP and semantic-matching workflows using transformer embeddings and Hugging Face models for entity normalization, noisy-data matching, intent classification, and improved retrieval coverage.
- Integrated AI inference services with backend APIs, carrier systems, analytics dashboards, web interfaces, React Native applications, and native mobile clients.
- Improved frontend, mobile, and backend reliability through caching, batching, structured logging, automated testing, crash monitoring, performance profiling, A/B testing, and production feedback loops.
- Investigated production issues across frontend, backend, mobile, AI, data, and infrastructure layers, coordinating fixes and validating releases across web, iOS, and Android environments.
- Shipped and operated full-stack and AI services using Docker, CI/CD pipelines, cloud infrastructure, and observability tooling while balancing reliability, latency, infrastructure cost, and release stability.

Software Consultant · Fast Enterprises, Centennial

JAN 2018 — OCT 2019

- Developed and maintained backend applications using C#, .NET, and Microsoft SQL Server, with an emphasis on scalability, maintainability, data integrity, and operational reliability.
- Implemented database-driven business logic and workflow automation that reduced manual processing and improved internal efficiency.
- Built fraud-scoring models for tax-return data using historical datasets and statistical methods to identify anomalous patterns for further investigation.

Software Engineering Intern · ArcBest Technologies, Fort Smith

MAY 2016 — AUG 2016

- Developed C# applications and database workflows for internal business teams, improving day-to-day operational tooling.
- Implemented automation that reduced manual effort and improved the reliability of internal processes.
- Added unit tests and technical documentation to improve maintainability and handoff readiness.

EDUCATION

Bachelor's Degree - Computer Science · University of Arkansas, Fayetteville

APR 2015 — MAY 2017